

each sensor to the central computer through CP2012 module, which is an integrated USB transceiver.

Data transmitted is stored in .csv (comma-separated values) format for ease of access. Bluetooth module HC05 is shown in Fig. 7.

CP2012 module is shown in Fig. 8. Here, CP2012 module is connected with pin-to-pin connector to transfer

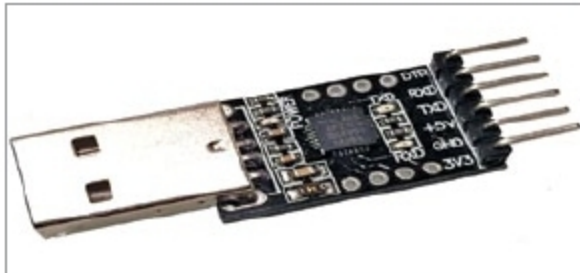


Fig. 8: CP2012 module



Fig. 9: Goodsky GU-SH-112D SPDT relay

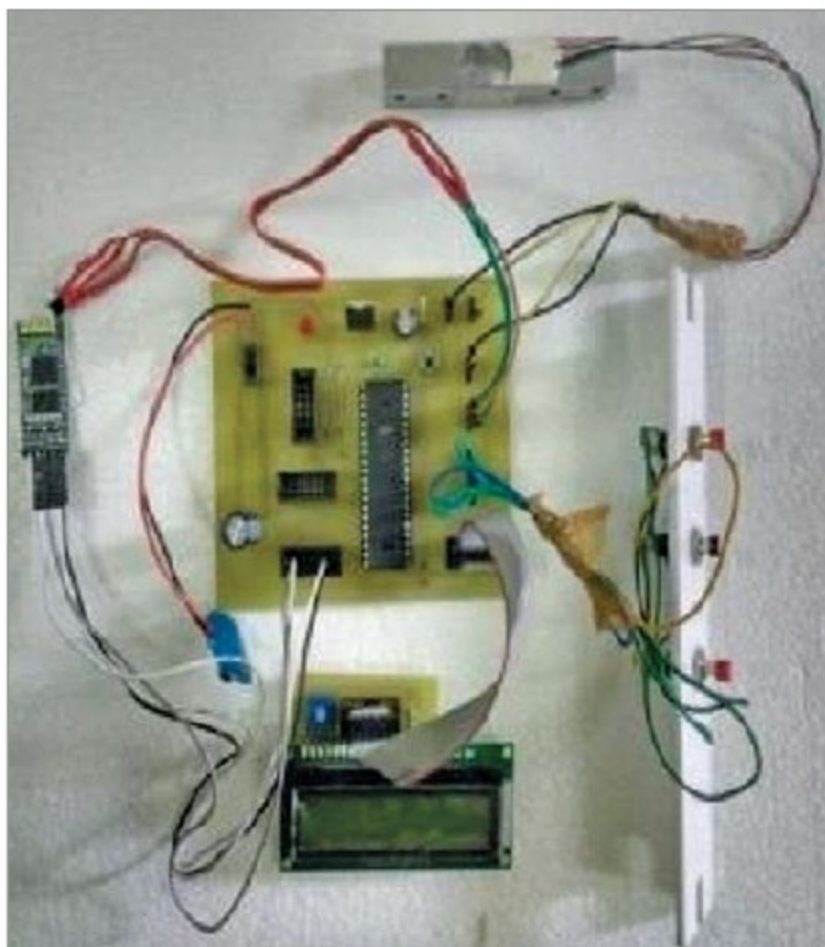


Fig. 10: Experimental setup of the server

data to the central computer.

Relay. A relay uses electromagnetic induction to connect or disconnect a contact between two terminals—it is like a mechanical switch but in autonomous mode. This relay is used for the central server. The relay used here is Goodsky GU-SH-112D, which is an SPDT relay operating at 12V DC. It can bear 30A current for 240V AC, which is sufficient for the central server. It is shown in Fig. 9.

Software used

The following software have been used for the system:

Python Scripting. OpenCV Python Scripting language is used for fingerprint reader access, authentication and recognition.

Android Studio. This is used for creating the graphical user interface (GUI) of the Android app.

PHP V5.3. PHP is a programming language used for designing the server. It plays an important role in sending SMSes.

Open Shift Cloud. Open Shift Cloud software is a platform for the service cloud. The server code needs to host somewhere and, hence, Open Shift Cloud is used for the same.

The prototype is prepared by integrating sensors, transceivers and Raspberry Pi. An experimental setup of the server is shown in Fig. 10, which represents the circuit of the smart fridge that uses ATmega16 MCU connected to Raspberry Pi using the Bluetooth module.

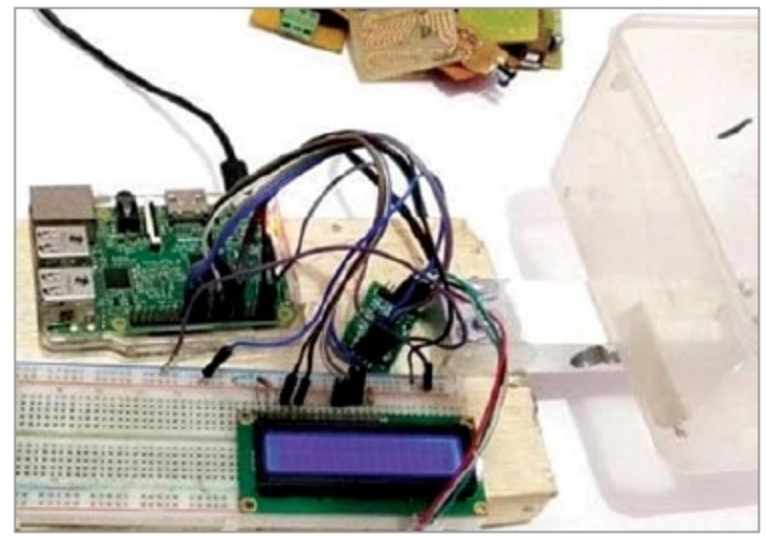


Fig. 11: Load cell connected to Raspberry Pi before testing

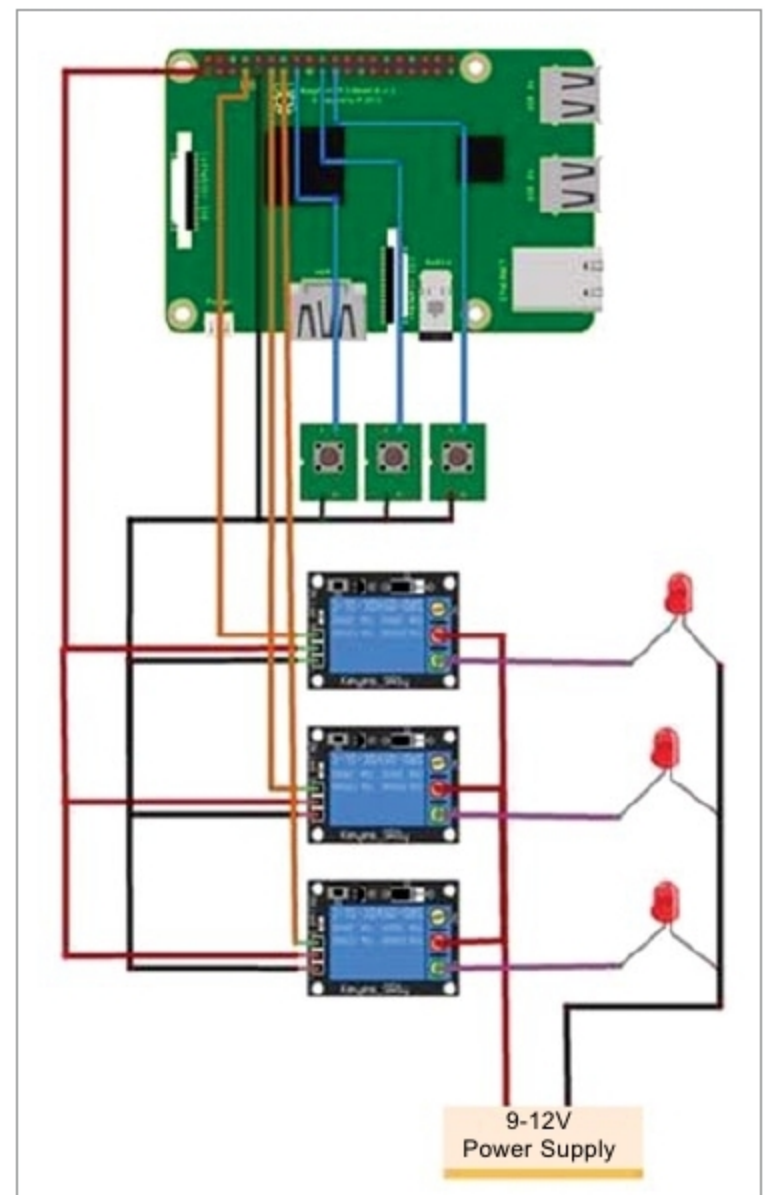


Fig. 12: Schematic view of the smart fridge

Push buttons are used for placing the order manually. The load cell is connected to Raspberry Pi as shown in Fig. 11, before testing. Schematic view of the smart fridge is shown in Fig. 12.

Home networking and architecture designs are important for a smart home automation system. Interfacing sensors and MCU with Raspberry Pi has been successfully simulated and tested for all smart nodes along with the remote-control feature using Android app. **EFY**